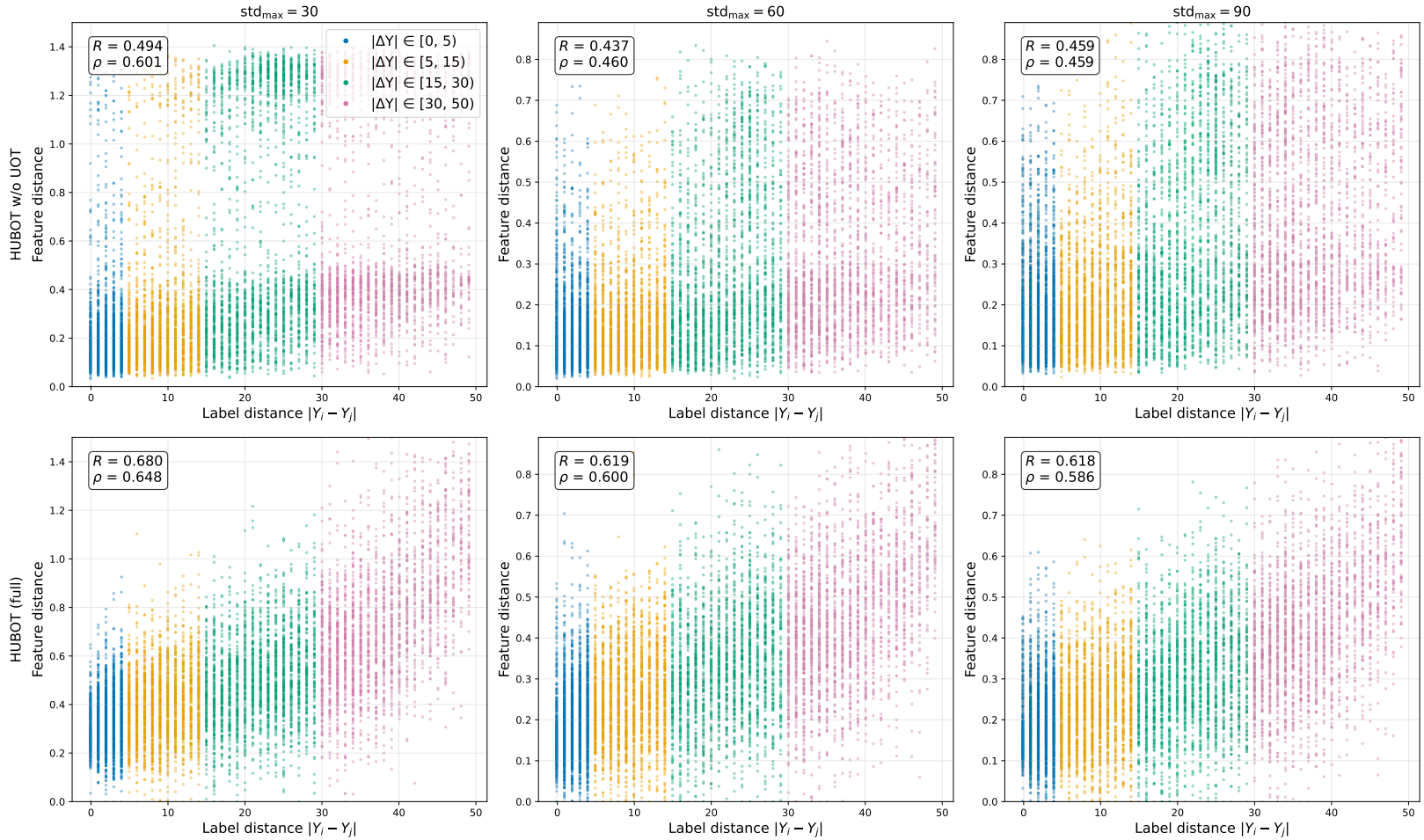

Figure 1: Feature distance vs. label distance on UTKFace (seed 1, last epoch) for **HUBOT w/o UOT (top row)** and **HUBOT full (bottom row)**, under three noise levels ($\text{std}_{\max} = 30, 60, 90$). Each point represents a pair of samples, colored by their label distance bin. Pearson R and Spearman ρ are reported in each panel. HUBOT full shows substantially higher correlation between feature distance and label distance, confirming that the UOT contrastive loss enforces a more order-preserving feature structure aligned with label proximity.